

Skye Gunasekaran

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SUMMARY

I am an AI researcher working toward systems that achieve more by using information better. I study how models can identify, preserve, and act on what matters while reducing unnecessary computation and complexity, with the broader goal of making high-performance AI more efficient, reliable, and legible.

SKILLS

Languages: Python, C/C++, Bash | **ML Stack:** PyTorch, HuggingFace Transformers, FlashAttention, flash-linear-attention | **Infra:** multi-GPU / distributed training (FSDP), Docker, Linux | **Focus:** LLM post-training, training infrastructure, efficient architectures, knowledge distillation.

EXPERIENCE

Research Intern

Jun. 2026 – Jul. 2026

ZeroEntropy (Acquired by Notion)

San Francisco, CA

Delivered ZeroEntropy's next generation 2B/4B rerankers (pending open-source release) **boosting recall@100 by 2.5 pts** vs prior generation and achieving SOTA on internal production benchmarks. **Re-engineered** training stack: migrated to Qwen 3.5, implemented **flash-linear attention**, and added sequence length bucketing, and additional training optimizations.

Developed a **context-preview system** that helps coding agents retrieve less redundant file content by serving summaries of elided spans through a Modal API (pending open-source release). On **DeepSWE** (GPT-5.6-Sol), held task-resolution parity (McNemar $p = 0.85$) while **cutting cumulative file-read content by 20%**, and net savings scaling with read intensity ($\rho = 0.30$, $p = 0.001$) above a **\$0.43/task** break-even.

Undergraduate Researcher

Jan. 2023 – Present

Neuromorphic Computing Group, UC Santa Cruz

Santa Cruz, CA

Independently led research on efficient transformer architectures, knowledge distillation, and uncertainty-aware forecasting, training models at scale on multi-GPU clusters. Produced four publications across Nature Communications, Nature Computational Science, APL Machine Learning, and ICLR, with contributions spanning algorithm design, large-scale experimentation, and training infrastructure.

Lead Tutor

Sep. 2024 – Dec. 2025

Baskin Engineering, UC Santa Cruz

Santa Cruz, CA

Developed and delivered lectures, assignments, and projects for a **150+ student ML research course**, coordinating a team of tutors and managing course logistics to support undergraduate research across multiple quarters.

RESEARCH

S. Gunasekaran, T. Wright, R.-J. Zhu, and J. K. Eshraghian, "Transformers with Selective Access to Early Representations," arXiv preprint arXiv:2605.03953 (2026). Under review at TMLR.

Reframed early-representation reuse as a **selective retrieval problem**: later layers learn when tokens and heads should recover first-layer information. Matched dense connectivity methods while using **31% less memory** and delivering **$\sim 1.8\times$ higher throughput**.

A. Kembay, **S. Gunasekaran**, R.-J. Zhu, Y. Zhang, and J. K. Eshraghian, "Efficient knowledge distillation via salient feature masking," *APL Machine Learning* **4**, 016104 (2026).

Asked whether a student learns better from a teacher's **most informative evidence**, rather than its full activation stream. A drop-in top- K masking rule improved accuracy by up to **6.39 points** across **15+ teacher-student pairs**.

S. Gunasekaran, A. Kembay, H. Ladret, R. J. Zhu, L. Perrinet, O. Kavehei, and J. Eshraghian, "A predictive approach to enhance time-series forecasting," *Nat. Commun.* **16**, 8645 (2025).

Led **Future-Guided Learning**, where hindsight teaches a forecasting model which errors deserve the largest updates. Improved seizure-prediction AUC-ROC by **44.8%** and reduced nonlinear forecasting error by up to **48.7%**; the article reached **37K+ accesses**.

R. J. Zhu, **S. Gunasekaran**, and J. Eshraghian, "Bridging the gap between artificial intelligence and natural intelligence," *Nature Computational Science* **4**, 559 (2024).

Co-authored a *News & Views* perspective on whether intelligence is best built from complex units or networks of simple ones, connecting biological fidelity to interpretability and neuromorphic efficiency.

S. Gunasekaran, J. Eshraghian, R. Zhu, and Z. Kuncic, "Knowledge distillation through time for future event prediction," in *The Second Tiny Papers Track at ICLR* (2024).

Introduced **knowledge distillation through time**: a future-observing teacher guides a student predicting from the present, establishing the idea that became Future-Guided Learning.

AWARDS & SERVICE

- **Baskin School of Engineering Commencement Speaker**, UC Santa Cruz (2026).
- NSF REU Fellowship, National Science Foundation (2025).
- Mantey Leadership Award, UC Santa Cruz (2025).
- Goldwater Nominee, UC Santa Cruz (2025).
- Dean's Award, Baskin School of Engineering, UC Santa Cruz (2024).
- Reviewer, ICLR 2026 Agents in the Wild Workshop (2026).
- Full Member, Sigma Xi, The Scientific Research Honor Society.

EDUCATION

University of California, Santa Cruz

Sep. 2022 – Jun. 2026

B.S. in Computer Science; GPA: 3.65

Santa Cruz, CA

Advisor: Prof. Jason K. Eshraghian, Neuromorphic Computing Group.